

**Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of
Selective Interference**

Jordan B Gunn

Department of Psychology, University of Cambridge

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Abstract

Visuospatial tasks performed after trauma-film encoding, or after a later reminder of the film, can reduce intrusive memories while leaving voluntary memory relatively intact. This selective interference effect is often treated as evidence that intrusive and voluntary remembering depend on differently vulnerable memory representations. Here we simulate an alternative retrieved-context account in an emotional Context Maintenance and Retrieval (eCMR) model, in which context reinstatement, competitor encoding, and retrieval control produce the selective-interference pattern without separately vulnerable representations. In the model, film context remains active after encoding and can later be reinstated by reminders, so an interfering task performed in either state associates task items with context states similar to film context. When later retrieval contexts cue both film items and task competitors, film items are less likely to be sampled during unguided retrieval, reducing intrusions even though film-item strength is not reduced. Deliberate recall is less vulnerable when task goals provide a film-directed start cue and selection favors film items over competitors; externally supplied film-related cues can also steer access toward film context across test formats. Together, the simulations reproduce selective impairment of unguided retrieval relative to deliberate recall and specify when the intrusion-voluntary-memory dissociation should be more or less pronounced, helping to interpret heterogeneity across empirical paradigms. Selective interference is therefore compatible with a single retrieved-context system and is not by itself diagnostic of separate representational systems or selective trace weakening.

Keywords: intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

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Introduction

Traumatic experience foregrounds a tension between durability and selectivity in episodic memory. Highly consequential events should be retained well enough in memory that critical details remain accessible long into the future. At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current cues, goals, and safety. Intrusive memories after trauma suggest that durable encoding can outpace regulatory control. The event returns without being deliberately sought and without being well matched to the demands of the present ([Brewin et al., 2010](#); [Ehlers & Clark, 2000](#); [Iyadurai et al., 2019](#); [Krans et al., 2009](#)). Understanding why such memories arise, and how they differ from voluntary memory for the same event, is therefore a shared target for both clinical science and basic memory theory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event ([Holmes & Bourne, 2008](#); [James et al., 2016](#)). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain available to deliberate retrieval while becoming more or less likely to return involuntarily. In such research, voluntary memory can be relatively preserved, and its relationship to intrusion frequency can be weak, such that the two measures dissociate ([Holmes et al., 2004](#); [James et al., 2016](#)). The same encoded episode can therefore support both involuntary and deliberate access, and these modes can come apart empirically ([Holmes et al., 2004](#); [Iyadurai et al., 2019](#)).

Visuospatial interference tasks make this dissociation experimentally sharper. When brief tasks such as Tetris are performed after film encoding, subsequent intrusions can be reduced while voluntary memory for the same material shows little measurable impairment ([Deepröse et al., 2012](#); [Holmes et al., 2009, 2010](#); [Lau-Zhu et al., 2019](#)). Related post-reminder studies show that a reminder-plus-Tetris procedure can reduce later intrusions after a longer delay ([James et al., 2015](#);

[Kessler et al., 2020](#)). Similar procedures have also been extended to real-life traumatic events, including motor vehicle accidents, emergency caesarean section, and war-related trauma ([Horsch et al., 2017](#); [Iyadurai et al., 2018](#); [Kanstrup et al., 2021](#)). At the same time, the evidence base is heterogeneous, and recent reviews and replications caution against treating the effect as settled or uniform ([Asselbergs et al., 2023](#); [Varma et al., 2024](#); [Wessel et al., 2025](#)). This selective interference effect therefore raises a theoretical question: when and why can a post-encoding or post-reminder manipulation alter one mode of retrieval more than another, even while both draw on the same recent experience?

One prominent family of interpretations treats the selective interference effect as evidence that the memory underlying intrusions is more vulnerable to interference than the memory supporting deliberate recall ([Brewin et al., 1996](#); [Brewin, 2014](#)). Dual-representation accounts make this claim architecturally: traumatic experience is proposed to produce distinct sensory and contextual memory traces, with the first supporting involuntary re-experiencing and the second supporting deliberate recall ([Brewin et al., 2010](#); [Brewin, 2014](#)). In immediate post-film interventions, a visuospatial task is then understood to compete with consolidation of the sensory or intrusion-supporting trace, reducing later intrusions while leaving the contextual trace and voluntary memory relatively preserved ([Deepröse et al., 2012](#); [Holmes et al., 2009, 2010](#)). For post-reminder interventions, related interpretations often appeal to reconsolidation or post-reactivation restabilization: the reminder reactivates the relevant trace, and the interfering task disrupts its later availability ([James et al., 2015](#); [Kindt et al., 2009](#)). Across these variants, selective interference is explained by treating the manipulation as acting more strongly on intrusion-supporting or reactivated memory than on the basis for deliberate recall.

Here we argue that selective interference need not be interpreted as evidence for differently vulnerable memory representations. It may instead reflect differences in retrieval processes operating over a shared set of episodic representations. Mechanisms supporting selective, goal-directed retrieval in voluntary recall may be sufficient to sidestep or overcome interference that nonetheless reduces the likelihood of unintentional recall. On this interpretation,

the selective-interference effect reflects a capacity to intentionally focus retrieval on goal-relevant event information while reducing the influence of competing memories.

We formalize and evaluate this proposal in an emotional Context Maintenance and Retrieval model (Cohen & Kahana, 2022; Polyn et al., 2009a; Talmi et al., 2019).

Retrieved-context models explain recall by treating context as a gradually changing state that is updated by studied items, reinstated by recalled items, and used to cue later memory search (Howard & Kahana, 2002; Polyn et al., 2009a; Sederberg et al., 2008). CMR extends this framework by representing context-to-item and item-to-context associations across temporal, semantic, and source/contextual dimensions (Polyn et al., 2009a, 2009b). Control over memory search enters the framework in two related ways: start-of-recall context reinstatement can bias retrieval toward a target episode, while monitoring mechanisms can reject candidates that poorly match a target context (Lohnas et al., 2015; Sederberg et al., 2008). Emotional CMR variants (i.e., eCMR and CMR3) add emotional context and emotion-dependent binding, allowing retrieved-context models to capture how affective content shapes recall organization, emotional intrusions, and intentional access (Cohen & Kahana, 2022; Talmi et al., 2019). Together, these commitments make the framework a principled setting for demonstrating how selective interference can arise within a single affectively-modulated episodic-memory system.

For selective interference, our model treats the film, any reminder, and subsequent interference as episodes encoded by binding item representations to a continuously evolving context representation (Howard & Kahana, 2002; Polyn et al., 2009a; Yonelinas et al., 2019). Post-film or post-reminder activity can therefore encode task items in context states similar to film-associated context states, including states reinstated by reminders. At retrieval, the current context state cues memory search, so competition depends on how strongly film items and task items are associated with similar context states. Deliberate recall can reduce vulnerability to this competition when retrieval begins from a more film-directed state and when selection favors film items over competitors. The account therefore attributes the intrusion-voluntary-memory dissociation to retrieval modes and control settings applied to shared memory associations. This

reframes selective interference as a change in which competitors are cued by film-associated context states, rather than as reduced availability of film items after encoding or reminder reactivation.

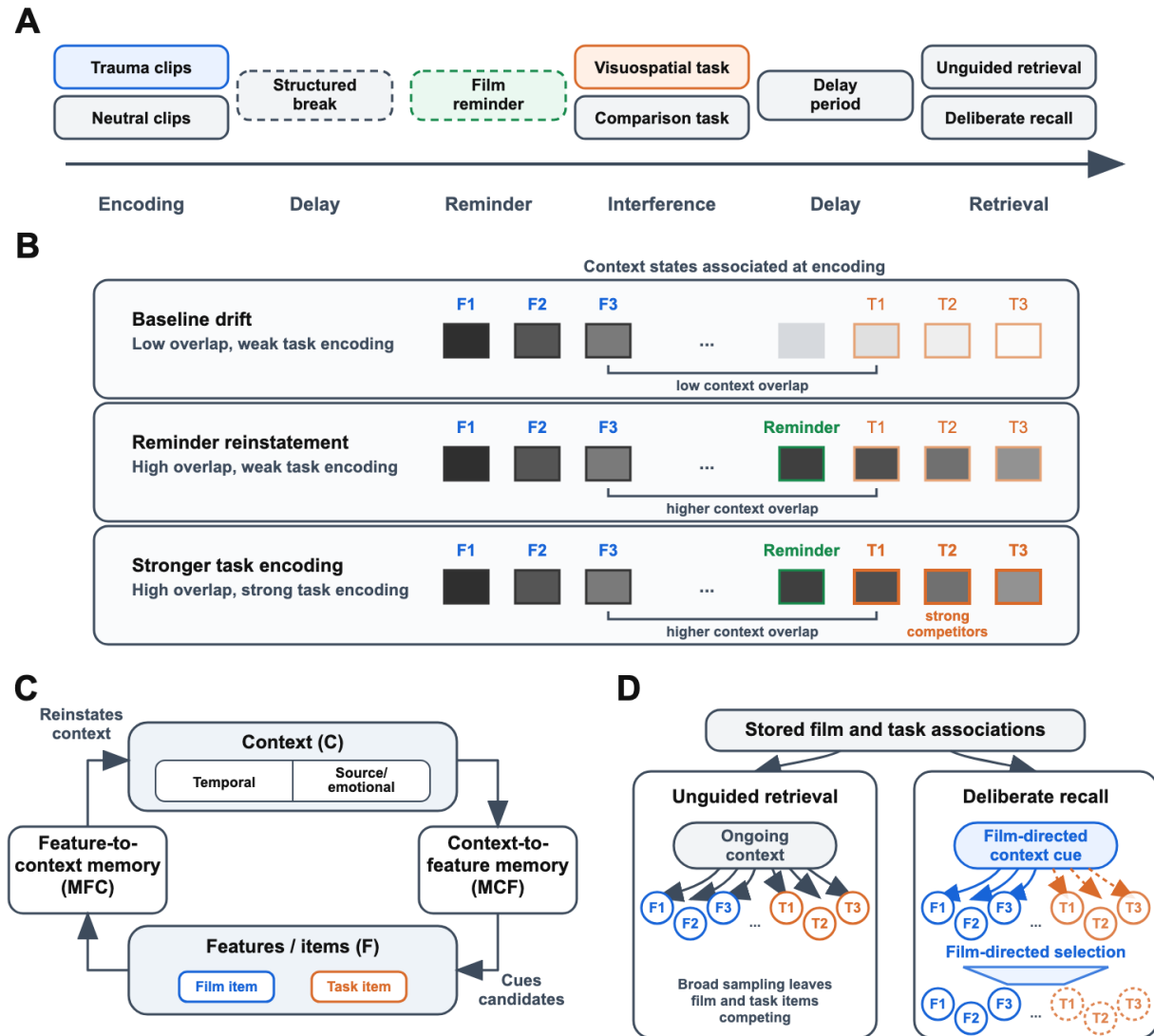
Our simulations evaluate the account at the level where a retrieved-context model makes its distinctive predictions: the distribution and organization of recall across the film, interference, and delay phases of a trauma-film paradigm. Simulation 1 demonstrates the context-binding account of interference: after a film reminder, a strongly encoded task can reduce later film recall because task items are encoded while context is still film-relevant, making them likely to compete when film context cues retrieval. Simulation 2 adds the intentionality manipulation, comparing unguided retrieval with deliberate recall after the same task manipulation and showing how film-directed recall can reduce the impact of task competitors. A later task-specificity simulation will clarify which task properties matter in the model, including task-item learning strength and overlap with film-associated context, rather than treating the visuospatial/comparison contrast as primitive. Simulation 4 examines cue support across test formats, showing how film-related monitoring cues or supplied film items can reduce reliance on broad context-to-item sampling and thereby narrow the intrusion-voluntary-memory dissociation ([Lau-Zhu et al., 2021](#)). The result is a mechanistic alternative to separately vulnerable memory representations and a set of specific hypotheses about when selective interference should appear and how it should vary across paradigms.

A Retrieved-Context Account of Selective Interference

The simulations rely on three retrieved-context mechanisms. Context-to-item retrieval uses current context to cue candidate items. Competition follows when film items and task competitors are associated with similar context states. Item-to-context retrieval allows recalled items, reminder cues, and supplied film items to reinstate associated context. Film-directed retrieval control changes the starting cue and selection rule used to favor film items over task competitors. Figure 1 shows how these mechanisms map onto the trauma-film paradigm.

Figure 1

A retrieved-context account of selective interference. The account explains selective interference through context overlap, task encoding strength, and retrieval control within a shared memory system. **(A)** The simulated paradigm includes film encoding, a structured break, a film reminder, interference or comparison task, a delay period, and retrieval tests. **(B)** Film and task items remain distinct while becoming associated with context states active at encoding; contextual overlap makes task items likely to be cued alongside film items, and task encoding strength determines how strongly they compete once cued. **(C)** Context-to-item associations cue candidate items, and item-to-context associations let recalled, reminded, or supplied items reinstate associated context. **(D)** The same stored associations support unguided retrieval and deliberate recall, with deliberate recall gaining protection from a film-directed start cue and film-directed selection.



Contextual similarity is established during encoding. Film items and task items are distinct item representations, but each is linked to the context state active when it is encountered. Task items are most likely to share film-relevant context when they are encoded soon after the film, after a reminder has reinstated film-associated context, or under matching affective conditions. Delay and filler activity move context away from the film, reducing the likelihood that later retrieval cues film items and task competitors together. These conditions determine whether task items are likely to be cued alongside film items; task encoding strength determines how strongly they compete once cued. Emotional extensions of CMR represent affect as an additional context dimension (Cohen & Kahana, 2022; Talmi et al., 2019). In the present model, this affective-context stream can amplify competition when film and task items share the relevant affective feature.

Context-to-item retrieval uses the current context state to cue candidate items. Here, unguided retrieval refers to broad context-to-item sampling with little film-directed guidance from task goals or external cues. When a retrieval context resembles states associated with both film items and task items, the same cue supports both sets. Such broad, weakly guided retrieval is especially vulnerable to this competition: as task competitors receive support, any given film item becomes less likely to be sampled as an intrusion. The size of this effect depends on the number of competitors, their learned support, and the similarity between their associated temporal or affective context states and film context. Perceptual modality contributes through those quantities: a visuospatial task is effective to the extent that it creates competitors with sufficient learned support or sufficiently similar associated context.

Item-to-context associations allow recalled items, reminder cues, and supplied film items to reinstate associated context. In ordinary recall, a retrieved item shifts context toward the state in which that item was encoded, making nearby studied items more likely to be retrieved next. This context-updating loop is the standard retrieved-context explanation of temporal contiguity and temporally organized recall across filled delays (Howard & Kahana, 2002; Kahana, 1996; Sederberg et al., 2008). The same dynamics allow recent but irrelevant experiences to intrude during recall of a target episode, with intrusion frequency decaying gradually over intervening

time ([Howard et al., 2008](#); [Lohnas et al., 2015](#); [Zaromb et al., 2006](#)). In delayed-interference paradigms, a reminder of the film is used before task encoding in studies that interpret the procedure as memory reactivation ([James et al., 2015](#); [Kessler et al., 2020](#)). On the present account, the reminder uses the same item-to-context pathway to reinstate film-associated context, paralleling temporal-context explanations of human post-reminder effects ([Sederberg et al., 2011](#)). Delayed interference is therefore graded by reminder strength and subsequent drift, because those quantities determine how much film-associated context remains available when the task is encoded.

The same item-to-context pathway also matters at test. During recall or monitoring, retrieved or presented items update context as retrieval unfolds. Retrieving a film item pulls context toward film-associated states and can support additional film retrieval, whereas retrieving a task competitor pulls context toward task-associated states and can prolong competition. Film-related cues supplied during test can likewise update context, making subsequent retrieval less dependent on internal cue maintenance. Furthermore, retrieval tests that supply a film item can be less vulnerable to interference, because the supplied item can reinstate its associated context directly, sidestepping the problem of selecting between items associated with overlapping context states ([Lau-Zhu et al., 2021](#)).

Deliberate recall is protected only to the extent that retrieval control changes the search. Deliberate recall using context-to-item retrieval remains exposed to competitor interference. The first control lever is cue construction or start-state control, which reinstates film-associated context before recall and biases search toward film items. The second is competition focusing or filtering, which sharpens selection among candidates and favors strongly supported film items over weaker or off-target competitors. These levers yield graded protection: deliberate recall can lose film items when competition is strong, but endogenous film-directed control can reduce that loss relative to broad retrieval from ongoing context.

The simulations instantiate these commitments in an emotional CMR model with phase-specific encoding, reminder-driven reinstatement, retrieval-mode and control settings, and

affective overlap (Cohen & Kahana, 2022; Talmi et al., 2019). The model departs from standard list-learning applications of CMR in two ways. First, film, break/delay, post-film task, and filler/delay periods form a continuous encoded stream, and a film reminder can reinstate film-associated context before task encoding. Second, non-film events from the break, task, and filler phases are encoded as distinct recallable items that can compete with film items during retrieval. This setup lets selective interference appear as a redistribution of retrieval across encoded phases, rather than as a direct reduction in film-item strength.

Model Specification

The specification below defines the model’s representations, parameters, and initialization before turning to context updating, associative learning, retrieval, retrieval-control settings, and affective context.

Representations and Parameters

Each study event i has an orthonormal feature vector $f_i \in \mathbb{R}^N$, where N is the number of encoded item features in the simulated paradigm. The model maintains a unit-length context vector $c_i \in \mathbb{R}^D$, where D is the number of context units. The core context units represent the start state and item-associated temporal context. Before encoding begins, context is initialized to the start state c_0 , with the start unit active and item-associated temporal-context units inactive. Subsequent events update this state, so the context associated with each film, task, or filler item depends on where it occurs in the continuous paradigm stream. The model contains two associative matrices. The matrix $M^{FC} \in \mathbb{R}^{D \times N}$ maps feature vectors to context input. The matrix $M^{CF} \in \mathbb{R}^{N \times D}$ maps context to feature support. All vectors are treated as column vectors, so $M^{FC} f$ is a context vector and $M^{CF} c$ is a feature-support vector.

Table 1 lists the symbolic base parameters used by the model. The equations below define the model architecture; numerical parameter values and phase-specific settings are specified in the simulation methodology.

Pre-experimental associations represent background memory structure already present before the simulated paradigm begins. They give each item baseline self-support and shared

support, so later retrieval reflects both pre-existing structure and associations learned during the film, task, and filler phases. Because the selective-interference simulations vary the new associations formed across phases, these baseline associations are treated as fixed model architecture rather than as the mechanism of interest. For the start unit and item-associated temporal context units:

$$M_{pre}^{FC}(j, i) = \begin{cases} 1 - \gamma & \text{if } j = i + 1 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$M_{pre}^{CF}(i, j) = \begin{cases} 0 & \text{if } j = 0 \\ \delta & \text{if } j = i + 1 \\ \alpha & \text{if } j > 0 \text{ and } j \neq i + 1 \end{cases} \quad (2)$$

The start context has no pre-experimental context-to-feature support. For context units corresponding to items, δ gives each item self-support and α gives all other items shared support. This initialization supplies baseline support before experimental learning.

Context Updating, Learning, and Retrieval

The same feature-to-context operation updates context when an item is encoded, recalled, or supplied as a cue. When feature f_i is active, it probes feature-to-context memory to retrieve context input:

$$c_i^{\text{IN}} = M^{FC} f_i \quad (3)$$

The model normalizes this input to unit length, yielding $\hat{c}_i^{\text{IN}}$. Context then drifts toward the normalized input with a drift rate β :

$$c_i = \rho_i c_{i-1} + \beta \hat{c}_i^{\text{IN}} \quad (4)$$

Let $d_i = c_{i-1} \cdot \hat{c}_i^{\text{IN}}$. The normalization coefficient is:

$$\rho_i = \sqrt{1 + \beta^2(d_i^2 - 1)} - \beta d_i \quad (5)$$

The coefficient ρ_i is the nonnegative solution that keeps c_i at unit length. When a new event is encoded, the active feature and updated context are also associated via outer-product learning. Feature-to-context associations update as:

$$\Delta M^{FC} = \gamma c_i f_i^\top \quad (6)$$

A primacy function weights context-to-feature learning:

$$\phi_i = \phi_s e^{-\phi_d \cdot i} + 1 \quad (7)$$

Context-to-feature associations update as:

$$\Delta M^{CF} = \lambda \phi_i f_i c_i^\top \quad (8)$$

Here, λ scales context-to-feature learning, while ϕ_s and ϕ_d define the size and decay of the primacy boost. During context-guided retrieval, current context cues encoded items through M^{CF} :

$$a_i = [M^{CF} c]_i \quad (9)$$

The choice sensitivity τ sharpens competition among candidates:

$$A_i = a_i^\tau \quad (10)$$

The probability of recalling item i , conditional on continuing recall, follows a Luce choice rule over currently unrecalled items:

$$P(i \mid \text{recall}) = \frac{A_i}{\sum_k A_k} \quad (11)$$

At each retrieval step, the model may terminate with probability $P_{\text{stop}}(j)$ or recall an item with probability $(1 - P_{\text{stop}}(j)) \cdot P(i \mid \text{recall})$. Termination probability grows with output position:

$$P_{\text{stop}}(j) = \theta_s e^{j\theta_r} \quad (12)$$

Here, j is the number of items already recalled, θ_s sets the baseline termination probability, and θ_r controls its growth. In simulations, this value is bounded to remain a valid probability. After a recalled item r_j , context is updated through the same feature-to-context pathway:

$$c_{j+1} = \rho_{\text{rec},j} c_j + \beta_{\text{rec}} \widehat{M^{FC} f_{r_j}} \quad (13)$$

This recalled-item update drives context toward the recalled item's encoding neighborhood, biasing subsequent retrievals toward nearby items and producing temporal organization in retrieval. Encoding operations use the context update and learning equations together, whereas reminders, recalled items, and supplied film items use the context update without adding a new recallable event.

Retrieval Control

Unguided retrieval and deliberate recall use the same retrieval-support and choice rules in Equation 9 and Equation 11. They differ in the control settings applied to the retrieval state and sampling process. The first deliberate-recall control setting is film-directed start/context cueing:

$$c_{\text{start}} = \rho_{\text{start}} c_N + \beta_{\text{start}} c_0 \quad (14)$$

The scale m_{start}^β controls the strength of this cueing in deliberate recall. The second control setting is sharper selection, implemented by scaling τ . Higher τ sharpens competition among all candidates. It protects film recall only when film-directed cueing gives film items stronger support than task competitors. The third control setting is target monitoring after a non-film item is sampled. Let $t_i = 1$ for film items and $t_i = 0$ otherwise, and define $\kappa_i = t_i + (1 - t_i)\kappa$. Target monitoring modifies the recalled-item context update in Equation 13. With target monitoring, the context update after sampled item i becomes:

$$c_{j+1} = \rho_{\text{rec},j} c_j + \kappa_i \beta_{\text{rec}} \widehat{M^{FC} f_i} \quad (15)$$

When $\kappa < 1$, sampled non-film competitors have less influence on the next context state than sampled film items. This convention is used for deliberate recall when simulations manipulate rejected-recall updating. Tests that supply a film item are represented as cue-driven context reinstatement using the same feature-to-context input operation as Equation 3:

$$c_{\text{film}}^{\text{IN}} = M^{FC} f_{\text{film}} \quad (16)$$

Supplied film items therefore use the same feature-to-context operation as recalled and reminded items.

Affective Context

The model also includes an affective-context pathway inherited from eCMR (Cohen & Kahana, 2022; Talmi et al., 2019). The affective pathway allows the simulations to examine whether shared affective state can amplify competition; it does not make task affect the explanation of selective interference. Each item has a two-unit affective feature vector s_i , with $s_i = [1, 0]^T$ for items assigned the marked affective feature and $s_i = [0, 1]^T$ for neutral or default items. The affective feature vector initializes M_{emot}^{FC} so that an item can cue its affective context components:

$$M_{\text{emot}}^{FC}(\cdot, i) = (1 - \gamma)s_i \quad (17)$$

Affective context-to-feature support begins at zero:

$$M_{\text{emot}}^{CF}(i, \cdot) = 0 \quad (18)$$

Thus, affective context can update during experience before it contributes retrieval support. Let $\phi_{\text{emot},i}$ denote the added context-to-feature learning strength for item i in the affective pathway. With the binary vector above, $\phi_{\text{emot},i} = \phi_{\text{emot}}[s_i]_1$, so items assigned the marked

affective feature receive ϕ_{emot} and neutral or default items receive 0. When affective learning is engaged, the affective components of context receive additional context-to-feature learning:

$$\Delta M_{\text{emot}}^{CF} = \lambda \phi_i \phi_{\text{emot},i} f_i c_{\text{emot},i}^{\top} \quad (19)$$

Here, $c_{\text{emot},i}$ denotes the affective components of context. Affective support enters retrieval through learned affective context-to-feature associations and contributes to the same competition defined in Equation 9. Task items are distinct item features and are assigned the marked affective feature only in affective-overlap variants. When film and task items share the marked affective feature and affective context-to-feature learning is enabled, Equation 19 can amplify competition beyond temporal proximity alone.

These equations define the item-context architecture used in all simulations. Because retrieval proceeds as a sequence of competitive samples terminated by a growing stop probability, task competitors can reduce film retrieval by winning individual retrieval competitions before search terminates.

Simulations

The simulations instantiate the retrieved-context account in a controlled trauma-film analogue. They vary phase settings and retrieval-control settings to examine how context overlap, task competition, and film-directed control shape recall. Simulation 1 isolates how task learning and film-context reinstatement create associations that make task items competitive during retrieval. Simulation 2 applies that competition to unguided retrieval and deliberate recall. Simulation 3 separates the control settings that can protect deliberate recall. Simulations 4–6 examine reminder timing, task specificity and affective overlap, and cue support across test formats.

Methodology

Base parameters were anchored to an existing immediate free-recall dataset in which participants who reported concern about the war in Ukraine studied and recalled lists of war-related and neutral photographs (Dupertuys et al., 2026). The fitted dataset included 123

participants, four lists per participant, 492 immediate recall sequences, and 3,804 scored recall events. The fitted parameters anchor baseline emotional-picture free-recall dynamics before the selective-interference manipulations are applied as phase-specific encoding or retrieval-control settings. Table 2 gives the default numerical values used in the current simulations.

The current simulations update associations after context integration and disallow repeated recalls.

Each simulated trial contains 16 film items, 16 break/buffer items, 16 task items, and 16 filler items. Equal phase lengths make redistribution across film, task, and filler periods directly comparable. Retrieval allows up to 48 recall attempts. All encoded film, break/buffer, task, and filler items can enter the retrieval competition unless a later simulation explicitly changes the test format.

Film, break/buffer, task, and filler items are encoded by the same CMR operations defined above. Phase settings control temporal drift and context-to-feature learning strength. In the no-reminder branch used for Simulation 1, the reminder-start and film-item reminder-cue drift scales are set to zero. In the with-reminder branch, context first shifts toward the start state and then receives sequential film-item reminder cues before task encoding. Reminder cues update context through Equation 3 and Equation 4, but they do not apply Equation 6 or Equation 8 and do not add new recallable items. Table 3 summarizes the phase structure and settings used for the current Figure 2 simulations.

For each condition and sweep value, the model encodes the phase sequence, enters retrieval, samples recallable items using Equation 11 and Equation 12, and repeats this process across simulated trials. Serial-position curves estimate the probability that each encoded position appears at least once in the recall sequence. The model-state analyses in Figure 2 quantify two quantities that explain the behavioral curves. The context-similarity analysis computes cosine similarity between unit-normalized temporal context states associated with film-item encoding and temporal context states active during task-item encoding. The support-allocation analysis forms a film-context cue from the centroid of film encoding context states, applies that cue to

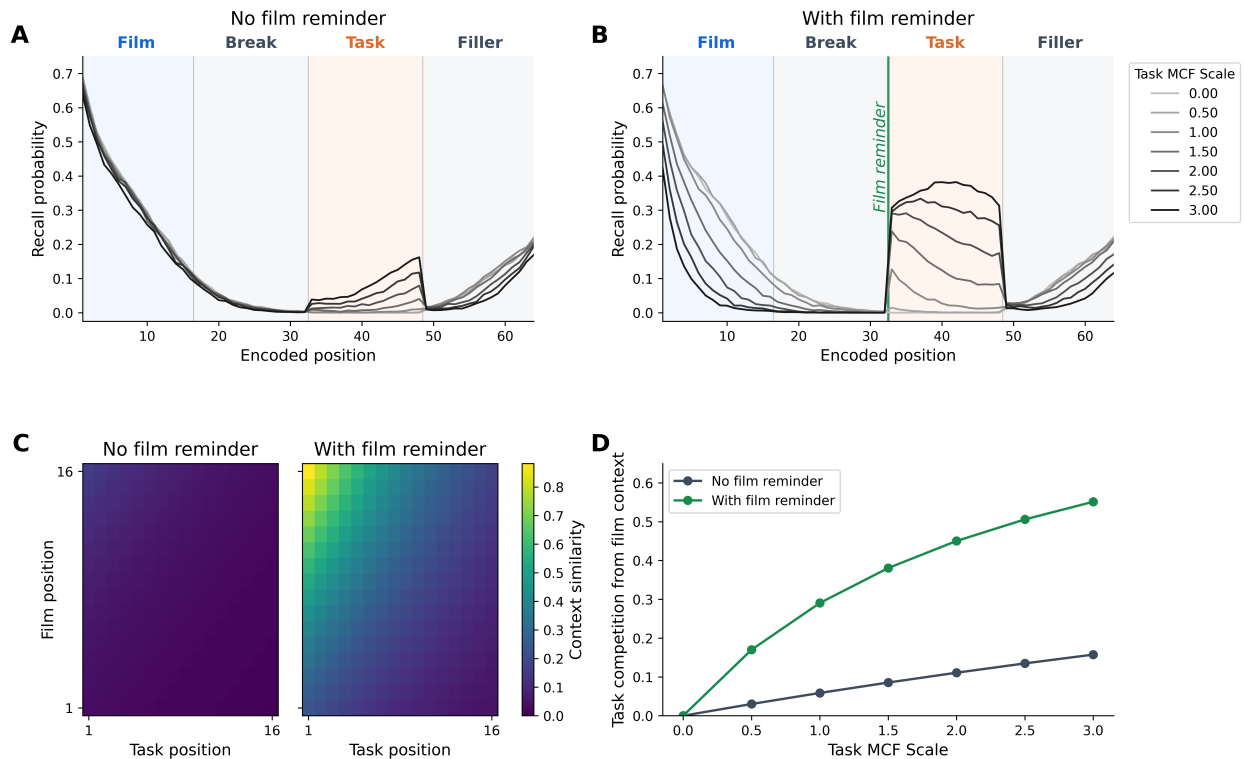
M^{CF} after task encoding, and computes task support divided by the sum of film and task support.

Simulation 1: Task Competition and Contextual Overlap

Simulation 1 isolates the conditions under which task learning strength becomes consequential for film recall. It sweeps task context-to-feature learning across the no-reminder and with-reminder branches described above. Retrieval settings are held fixed so Simulation 1 isolates the stored-association conditions under which distinct task items become effective competitors for film recall. Simulation 2 then applies those film-task associations to retrieval modes that differ in film-directed control.

Figure 2***Reminder-driven context overlap makes task encoding strength consequential for film recall.***

Simulation 1 demonstrates how task items can become competitors for film recall by varying task encoding strength with or without film-context reinstatement before task encoding. (A) Without a film reminder, task items become associated with later context states reached after drift away from the film, so stronger task encoding produces only modest redistribution from film items to task items. (B) With a film reminder, task items become associated with context states more similar to those associated with film items, so stronger task encoding produces larger redistribution from film items to task items. (C) This behavioral contrast follows from the context trace: reminder cues make the context states active during task encoding more similar to the states associated with film items. (D) When the model is cued with film context, stronger task encoding assigns a larger share of item support to task items, especially after a film reminder.



Panels A and B show phase-resolved recall probability across encoded positions. Without a film reminder, increasing task learning produces only modest loss of film recall and only modest

gain in task recall, because task items become associated with later context states reached after drift away from the film. With a film reminder, the same task-learning manipulation produces a much larger redistribution from film positions to task positions. The reminder branch therefore shows that task strength matters most when task items are associated with film-relevant context states. The serial-position curves also show that the effect is not uniform across film positions: early film recall remains relatively protected under the current settings compared with later film positions.

Panels C and D provide model-state analyses that explain why the behavioral curves diverge. Panel C quantifies temporal-context similarity between the states associated with film items and the states active during task encoding. When a reminder reinstates film-associated context before task encoding, the model encodes task items in states more similar to those associated with film items. Panel D quantifies support allocation from a film-context cue: as task learning increases, task items receive a larger share of item support when film contexts cue retrieval, especially after the film reminder. The resulting impairment is therefore not a weakening of film-item storage. It is an additive competition effect in which task items remain distinct from film items but become increasingly likely to win retrieval from film-relevant cues. Simulation 2 next applies these film-task associations to the retrieval-mode contrast needed for selective interference.

Simulation 2: Retrieval Mode and Selective Interference

Premise: Given stored film and task associations capable of producing competition, the second simulation asks whether unguided retrieval is more impaired than deliberate recall. The section should compare weak or absent task competition with strong task competition under unguided retrieval and deliberate recall. This is the first direct simulation of the selective-interference dissociation between intrusion-like access and voluntary recall.

Primary readouts: The main figure should show phase-resolved recall so the reader can see where retrieval probability is redistributed across film, task, and filler phases. A compact summary should also show film-item access and impairment from baseline for each retrieval

mode. Task-competitor recall should be reported or visible enough to show that lost film access is competition-driven.

Intended text take-away: This section establishes the core dissociation result. The model can reproduce the qualitative intrusion-voluntary-memory dissociation without separately weakening film items for one measure. Deliberate recall may still be impaired; the claim is relative protection, not immunity.

Open specification: The deliberate-recall settings used here need to be fixed before this section is written as a result. Those settings should be strong enough to create a retrieval-mode contrast but simple enough that Simulation 3 can still isolate the separate control mechanisms.

Simulation 3: Deliberate-Recall Control Mechanisms

Premise: After Simulation 2 establishes relative protection for deliberate recall, this simulation asks which control settings produce that protection. Deliberate recall is not categorically protected. Its relative sparing depends on control operations that make retrieval more film-directed. This section should separate the contributions of film-directed start/context cueing, sharper selection through τ , and target-monitoring or rejected-recall updating through κ if that mechanism remains in the simulation package.

Primary readouts: The figure should show how each control lever changes film-item access and the phase-resolved organization of recall under a fixed task-competition setting. The most useful contrast should include total film recall, task-competitor retrieval, and how closely recall remains tied to the film phase.

Intended text take-away: This section earns the “relatively intact voluntary memory” claim. Sparing is a graded consequence of retrieval control rather than a property attached to voluntary tests as a category. Weakening these control operations should make deliberate recall resemble unguided retrieval more closely.

Open specification: Decide whether target-monitoring through κ is a core control mechanism or a sensitivity analysis. The introduction currently emphasizes film-directed cueing and sharper selection more strongly than rejected-recall updating.

Simulation 4: Reminder and Timing Boundary Conditions

Premise: After Simulation 1 shows that reminder presence increases task competition, this simulation asks how delayed interference depends on graded reinstatement and drift. The delayed-reminder case applies the overlap mechanism after context has moved away from the film. A reminder reinstates film-associated context before task encoding by first shifting toward start context and then applying sequential film-item cues without new learning.

Primary readouts: The figure should show how weak versus strong reminder reinstatement changes phase-resolved recall and film-item access after a fixed task phase. If delay or filler length is varied, the figure should make clear whether the effect comes from reinstatement strength, subsequent drift, or their combination. Task-competitor retrieval should remain visible enough to show whether boundary-condition effects operate through competitor recruitment.

Intended text take-away: This section connects the model to post-reminder interventions. Delayed effects depend on graded reinstatement and drift, rather than an all-or-none storage-disruption window. The reminder restores the contextual condition required for later task encoding to affect film-item retrieval.

Open specification: Decide whether the main comparison is zero/weak reminder reinstatement versus strong reminder reinstatement, a reminder-strength sweep, a delay/drift sweep, or a compact factorial combination. Decide whether immediate interference remains a baseline within this section or is treated as the contrast condition established in Simulation 1.

Simulation 5: Task Specificity and Affective Overlap

Premise: This simulation asks what makes a visuospatial task effective in model terms. Competitor strength is one route by which task items become effective competitors. Contextual and affective overlap can amplify that competition when task items are cued alongside film items. This section should distinguish model-level task properties from the surface label of a task as visuospatial.

Primary readouts: The figure should show film-item access and task-competitor retrieval under task-strength and affective-overlap variants. If contextual overlap is also varied here, the

display should make clear which factor changes competitor recruitment and which factor changes competitor success once cued. Phase-resolved recall remains useful because the model's explanation depends on task items entering retrieval alongside film items.

Intended text take-away: The account does not require a special trace-weakening effect of visuospatial tasks. Task effectiveness can be decomposed into retrieved-context properties: competitor strength, contextual overlap, affective overlap, or their combination. Affective matching can be used as a sensitivity extension if it is run as a specific manipulation, but it should not become the main explanation unless the simulations make it central.

Open specification: Decide whether affective overlap is a main simulation or a sensitivity analysis. Decide whether comparison-task conditions are best represented as weak task learning, lower overlap, lower affective overlap, or a particular combination tied to an empirical contrast.

Simulation 6: Cue Support and Test-Format Extensions

Premise: Intrusion diaries, vigilance-style monitoring, free recall, cued recall, and recognition-like tests differ in how much they rely on broad context-to-feature search versus film-directed retrieval support. This section asks how film-related monitoring cues and tests that supply a film item alter vulnerability to task interference. The section should only become a simulation result if each test format is assigned a concrete retrieval or cueing setting. Candidate settings include unguided retrieval from ongoing context, monitoring with film-related cues, deliberate recall with film-directed start cueing, cued recall with stronger external support, and supplied-film-item access.

Primary readouts: The figure should show a graded vulnerability profile across retrieval and cueing settings under the same stored associations. The useful result is a continuum of film-item impairment across broad unguided search, film-cued monitoring, deliberate recall, and supplied-film-item access. The display should avoid implying a full recognition decision model unless such a mechanism is added.

Intended text take-away: This section interprets heterogeneity across empirical paradigms without treating intrusion and voluntary memory as fixed computational kinds. The model

predicts a continuum of vulnerability depending on how much the task supports film-directed retrieval or supplies film items as cues. Cue-rich tests should narrow the intrusion-voluntary-memory dissociation because they reduce reliance on broad context-to-feature sampling.

Open specification: The mapping from empirical test format to model setting is the main unresolved piece. If the mapping remains qualitative, this material belongs in the discussion rather than as a numbered simulation. Decide whether supplied-film-item access is implemented strongly enough to justify splitting this section into a separate recognition-like simulation later.

General Discussion

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Table 1*Base parameters and theoretical roles.*

Symbol	Name	Role
β_{enc}	encoding drift rate	Context drift during study
β_{start}	start drift rate	Start-of-list or film-directed context cueing
β_{rec}	recall drift rate	Context drift after recall
γ	feature-to-context learning rate	Learning rate for M^{FC}
λ	context-to-feature learning scale	Baseline or phase-scaled learning rate for M^{CF}
α	shared support	Pre-experimental shared support in M^{CF}
δ	self support	Pre-experimental self-support in M^{CF}
ϕ_s	primacy scale	Initial boost to context-to-feature learning
ϕ_d	primacy decay	Decay rate of primacy boost
τ	choice sensitivity	Sharpness of retrieval competition
β_{emot}	emotional-context drift rate	Context drift in the emotional pathway
ϕ_{emot}	emotional learning scale	Emotional context-to-feature learning
κ	target-monitoring scale	Reduced context updating after sampled non-film competitors when target monitoring is enabled
θ_s	stop scale	Baseline stopping probability
θ_r	stop growth	Growth of stopping probability across output positions

Table 2

Default numerical parameter values used in the current simulations. Values are from a pooled eCMR fit to the immediate free-recall sequences from Dupertuys et al. (2026), with α fixed to zero.

Symbol	Value	Role
β_{enc}	0.440	Encoding drift rate
β_{start}	0.766	Start-context drift rate
β_{rec}	0.678	Recall drift rate
γ	0.468	Feature-to-context learning rate
δ	0.00022	Pre-experimental self-support
α	0	Pre-experimental shared support
ϕ_s	0.499	Primacy scale
ϕ_d	0.530	Primacy decay
τ	3.562	Choice sensitivity
β_{emot}	0.696	Emotional-context drift rate
ϕ_{emot}	0.130	Emotional learning scale
θ_s	0.0123	Stop scale
θ_r	0.372	Stop growth

Table 3

Phase structure and settings for the current Figure 2 simulations. Scale values multiply the corresponding base rates or learning strengths.

Phase or setting	Current value	Role
Film items	16	Target episode
Break/buffer items	16	Intervening non-film experience before reminder/task
Film reminder cues	16	Sequential film-item cues, no new learning
Task items	16	Potential competitors
Filler items	16	Post-task non-film experience
Maximum recall attempts	48	Upper bound on sampled retrieval outputs
Simulated trials per sweep cell	5000	Replications used for Figure 2 estimates
Break and filler drift/learning scales	1.0	Baseline non-film encoding settings
No-reminder drift scales	0.0	Omits reminder-start and reminder-cue reinstatement
With-reminder drift scales	1.0	Applies reminder-start and film-item cue reinstatement
Task M^{CF} scale	0.0–3.0 by 0.5	Sweep of task context-to-feature learning strength
Affective marking	Off	No affective-overlap manipulation in Simulation 1